

Experiment Report

Course: Computer Programming

Experiment 1: C++ Basics & Expressions

Class:

Name:

Student ID:

Instructor:

Date: 2026.5.21

Experiment 1: Basics & Expressions

I. Objectives and Content

This experiment aims to practice C++ basics, which are the chapters 2-3 of the textbook.

The detailed goals of this experiment are as follows.

- **Syntax Mastery:** Understand preprocessor directives, main function structure, and namespace usage.
- **Data Integrity:** Distinguish between variables and constants, and understand memory allocation for different data types.
- **Logic Development:** Master operator precedence, associativity, and the side effects of assignment expressions.
- **Precision Control:** Learn to handle data type conversions (implicit and explicit) and format console output for readability.

Complete the following 8 programming tasks. Ensure all code includes proper documentation and follows the specific technical requirements for each problem.

II. Preparation

1. Install Visual Studio Code.
2. Install Compiler.
3. Open a folder and terminal in Visual Studio Code.
4. Set up the project folder.
5. Create .cpp files.
6. Compile and run.

III. Project Results

Task 1: Basic Structure & Documentation

1. **Description:** Detailed descriptions of the code.

- 1) Use a Block Comment to display Author, Date, Description.

```
/*  
 * Author: XXX  
 * Date: 2026-05-21  
 * Description: This program displays the user's name, department, and  
 student ID on three separate lines.  
 */
```

- 2) Output name line.

```
// Output name line  
cout << "Name: XXX" << endl; // Displays the name
```

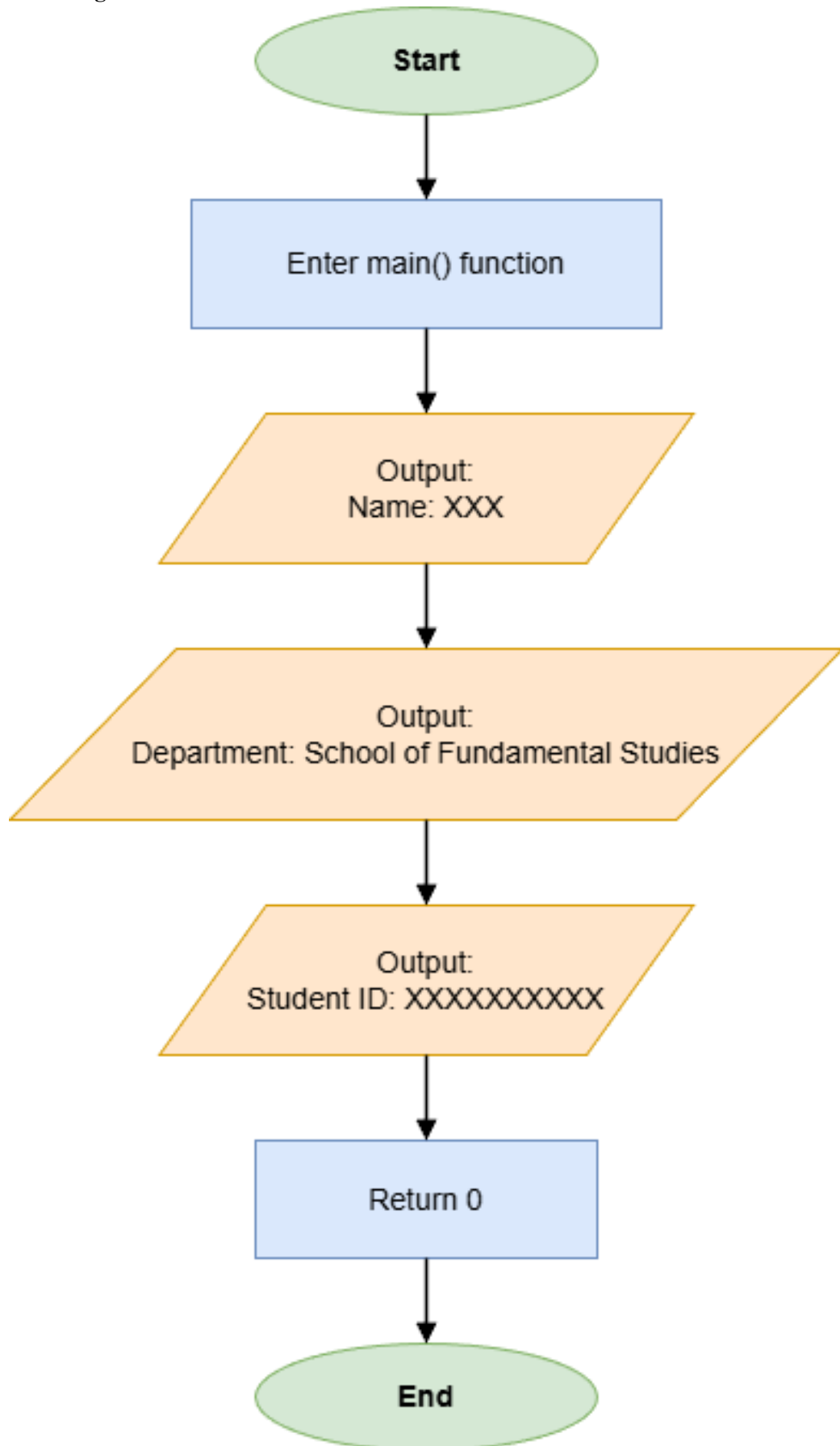
- 3) Output department line.

```
// Output department line  
cout << "Department: School of Fundamental Studies" << endl; //  
Displays the department
```

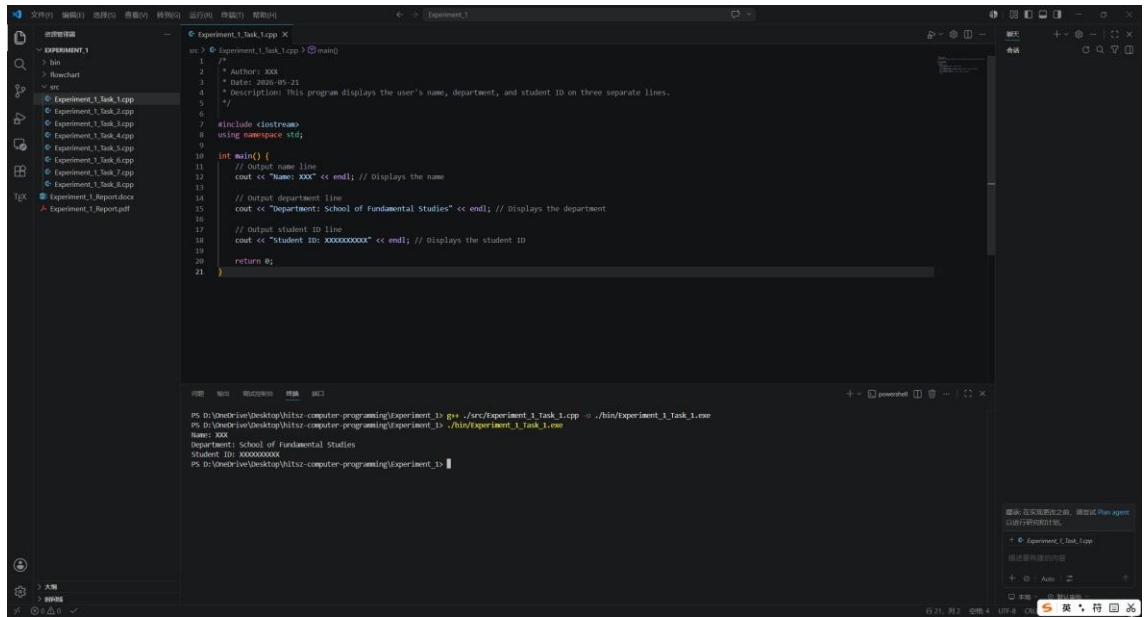
- 4) Output student ID line.

```
// Output student ID line  
cout << "Student ID: XXXXXXXXXX" << endl; // Displays the student ID
```

2. Flow Diagram:



3. Results (Output Screenshot): The screenshot of the output of the code.



The screenshot shows a C++ program being executed in a terminal window. The program, located at `Experiment_1_Task_1.cpp`, is a simple C++ file that prints the author's name, department, and student ID. The output of the program is displayed in the terminal window below the code editor.

```
1  /*  
2  * Author: XXX  
3  * Date: 2024-09-21  
4  * Description: This program displays the user's name, department, and student ID on three separate lines.  
5  */  
6  
7  #include <iostream>  
8  using namespace std;  
9  
10 int main() {  
11     // Output name line  
12     cout << "Name: XXX" << endl; // Displays the name  
13  
14     // Output department line  
15     cout << "Department: School of Fundamental Studies" << endl; // Displays the department  
16  
17     // Output student ID line  
18     cout << "Student ID: XXXXXXXXXX" << endl; // Displays the student ID  
19  
20     return 0;  
21 }
```

The terminal output shows the following results:

```
PS D:\Desktop\hitx-computer-programming\Experiment_1> g++ ./src/Experiment_1_Task_1.cpp -o ./bin/Experiment_1_Task_1.exe  
PS D:\Desktop\hitx-computer-programming\Experiment_1> ./bin/Experiment_1_Task_1.exe  
Name: XXX  
Department: School of Fundamental Studies  
Student ID: XXXXXXXXXX  
PS D:\Desktop\hitx-computer-programming\Experiment_1>
```

Task 2: Coin Value Calculator

1. Description:

- 1) Define each coin's value as a const unsigned int.

```
const unsigned int quarter = 25;
const unsigned int dime = 10;
const unsigned int nickel = 5;
const unsigned int penny = 1;
```

- 2) Use unsigned int for coin counts and unsigned long for the totalValue.

```
unsigned int numQuarters, numDimes, numNickels, numPennies;
unsigned long totalValue = 0;
```

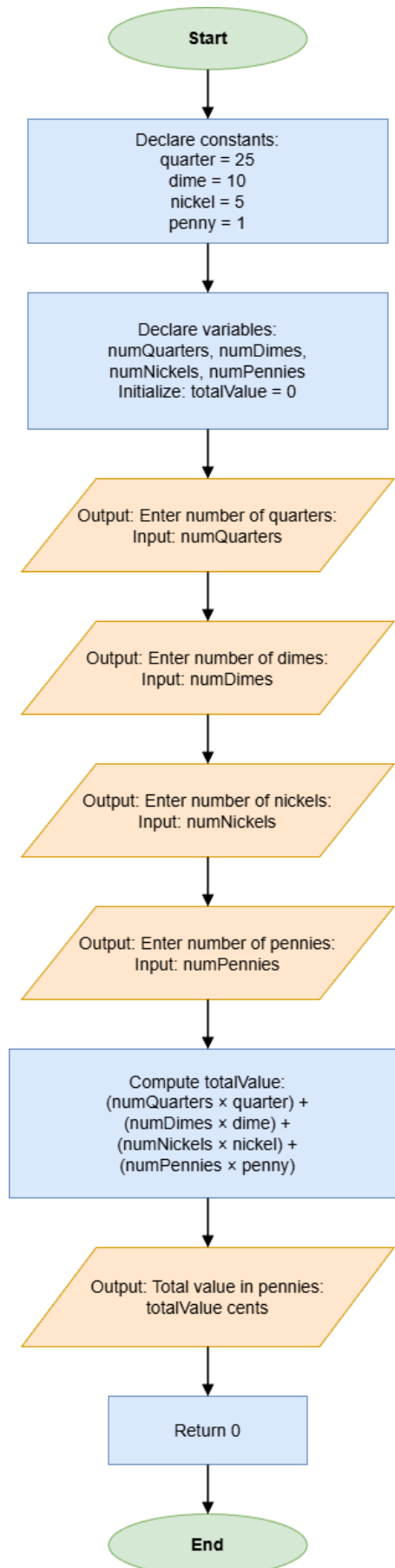
- 3) Input the number of quarters, dimes, nickels and pennies.

```
cout << "Enter number of quarters: ";
cin >> numQuarters;
cout << "Enter number of dimes: ";
cin >> numDimes;
cout << "Enter number of nickels: ";
cin >> numNickels;
cout << "Enter number of pennies: ";
cin >> numPennies;
```

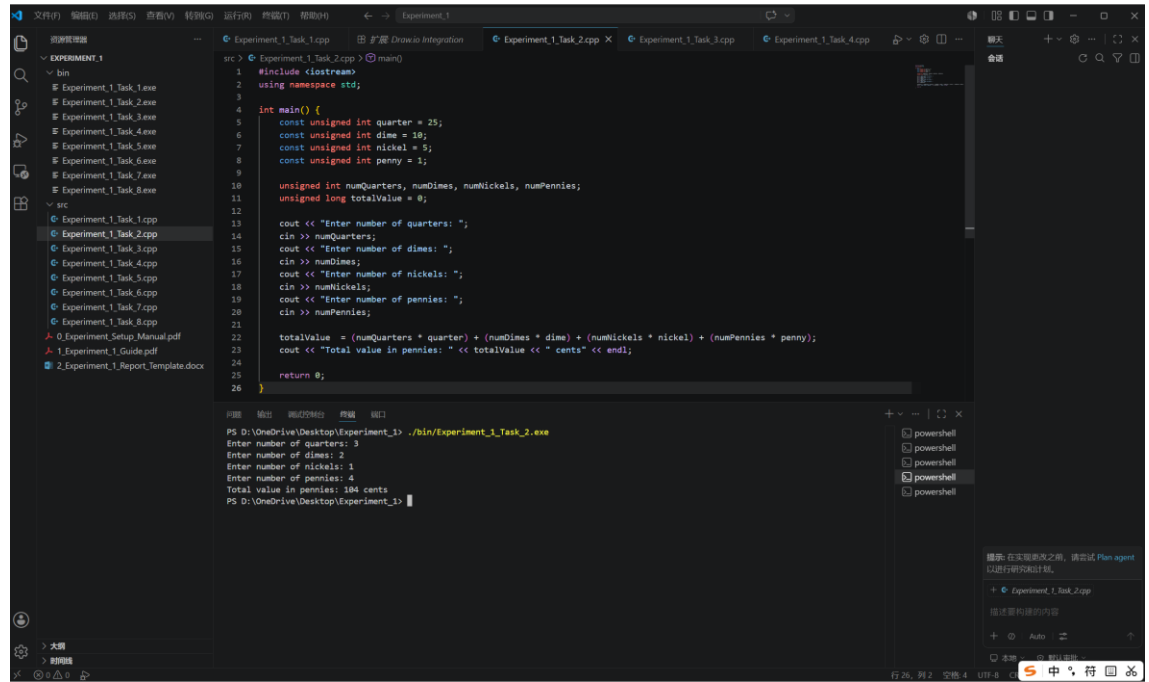
- 4) Calculate totalValue and print it.

```
totalValue = (numQuarters * quarter) + (numDimes * dime) +
(numNickels * nickel) + (numPennies * penny);
cout << "Total value in pennies: " << totalValue << " cents" << endl;
```

2. Flow Diagram:



3. Results (Output Screenshot):



```
src > Experiment_1_Task_2.cpp > main()
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     const unsigned int quarter = 25;
6     const unsigned int dime = 10;
7     const unsigned int nickel = 5;
8     const unsigned int penny = 1;
9
10    unsigned int numQuarters, numDimes, numNickels, numPennies;
11    unsigned long totalValue = 0;
12
13    cout << "Enter number of quarters: ";
14    cin >> numQuarters;
15    cout << "Enter number of dimes: ";
16    cin >> numDimes;
17    cout << "Enter number of nickels: ";
18    cin >> numNickels;
19    cout << "Enter number of pennies: ";
20    cin >> numPennies;
21
22    totalValue = (numQuarters * quarter) + (numDimes * dime) + (numNickels * nickel) + (numPennies * penny);
23    cout << "Total value in pennies: " << totalValue << " cents" << endl;
24
25    return 0;
26 }
```

```
PS D:\OneDrive\Desktop\Experiment_1> ./bin/Experiment_1_Task_2.exe
Enter number of quarters: 3
Enter number of dimes: 2
Enter number of nickels: 1
Enter number of pennies: 4
Total value in pennies: 104 cents
PS D:\OneDrive\Desktop\Experiment_1>
```

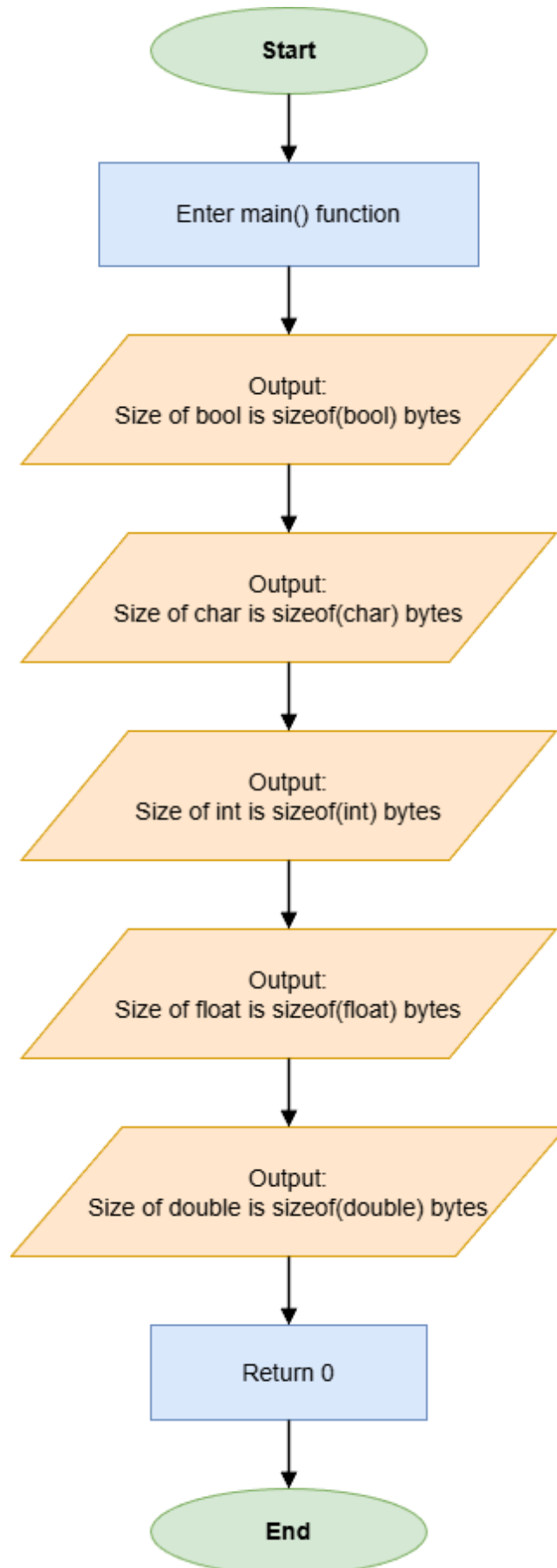

Task 3: Memory Size Analysis

1. Description:

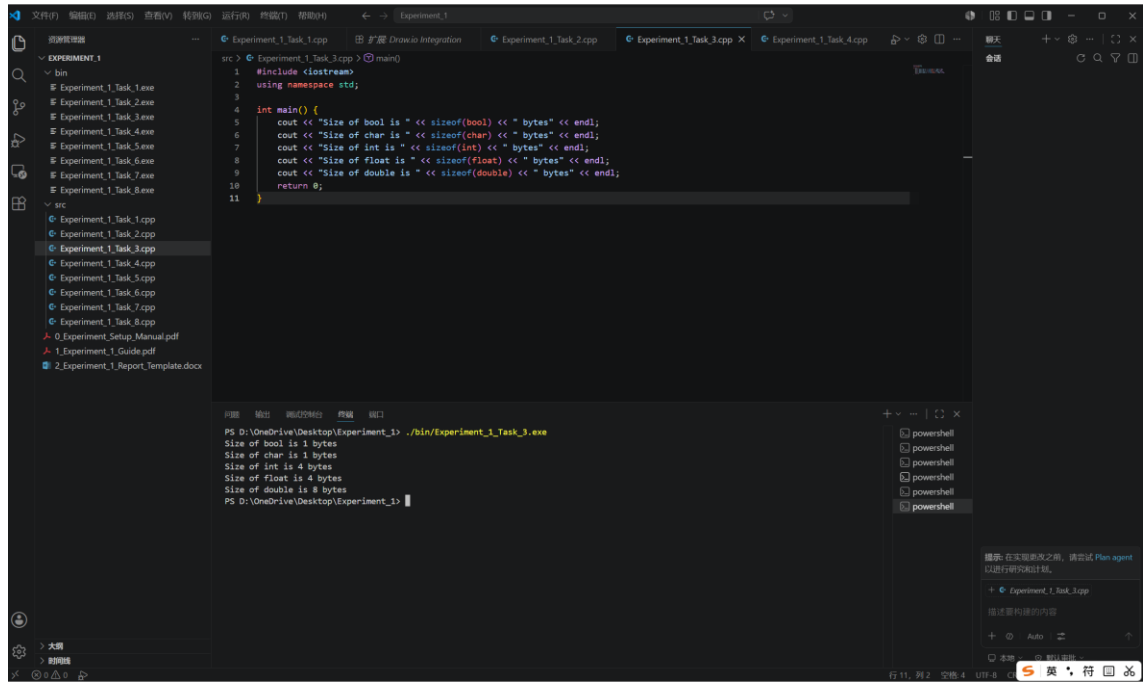
- 1) Use the sizeof operator to print the memory size of different data types.

```
cout << "Size of bool is " << sizeof(bool) << " bytes" << endl;  
cout << "Size of char is " << sizeof(char) << " bytes" << endl;  
cout << "Size of int is " << sizeof(int) << " bytes" << endl;  
cout << "Size of float is " << sizeof(float) << " bytes" << endl;  
cout << "Size of double is " << sizeof(double) << " bytes" << endl;
```

2. Flow Diagram:



3. Results (Output Screenshot):



The screenshot displays the Visual Studio Code interface. The left sidebar shows a file explorer with a project named 'EXPERIMENT_1' containing a 'bin' directory with several '.exe' files and an 'src' directory with several '.cpp' files. The main editor window shows a C++ file named 'Experiment_1_Task_3.cpp' with the following code:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     cout << "Size of bool is " << sizeof(bool) << " bytes" << endl;
6     cout << "Size of char is " << sizeof(char) << " bytes" << endl;
7     cout << "Size of int is " << sizeof(int) << " bytes" << endl;
8     cout << "Size of float is " << sizeof(float) << " bytes" << endl;
9     cout << "Size of double is " << sizeof(double) << " bytes" << endl;
10    return 0;
11 }
```

The bottom panel shows the output of the program, which is displayed in a PowerShell terminal window. The output is:

```
PS D:\OneDrive\Desktop\Experiment_1> ./bin/Experiment_1_Task_3.exe
Size of bool is 1 bytes
Size of char is 1 bytes
Size of int is 4 bytes
Size of float is 4 bytes
Size of double is 8 bytes
PS D:\OneDrive\Desktop\Experiment_1>
```

Task 4: Character and ASCII Interaction

1. Description:

- 1) Store the input in a char variable.

```
char character = 0;
```

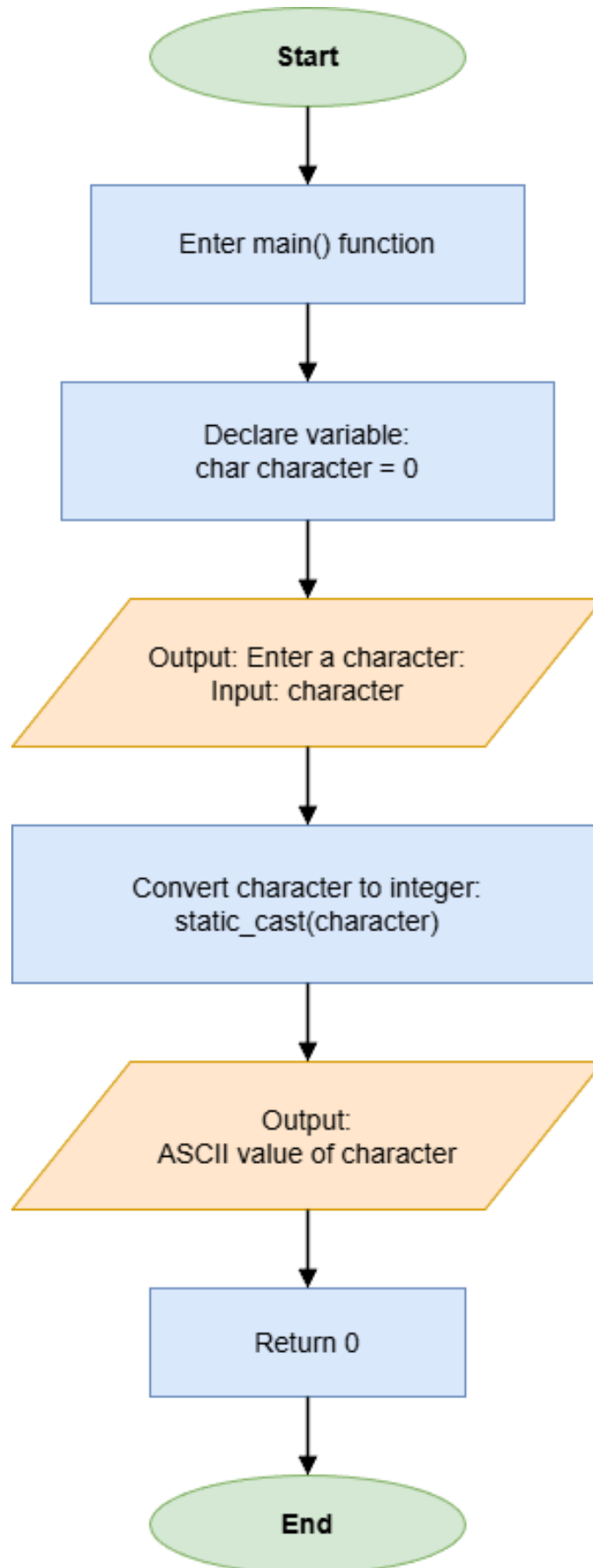
```
cout << "Enter a character: ";
```

```
cin >> character;
```

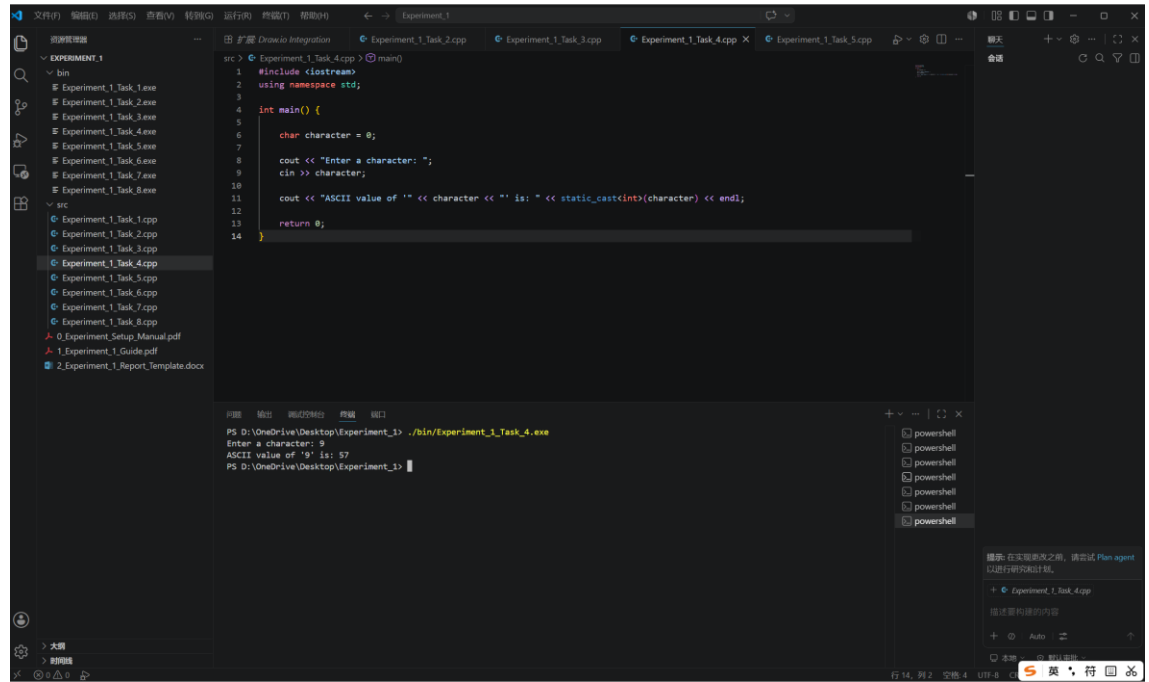
- 2) Display that character's decimal ASCII value using `static_cast<int>()`.

```
cout << "ASCII value of '" << character << "' is: " <<  
static_cast<int>(character) << endl;
```

2. Flow Diagram:



3. Results (Output Screenshot):



The screenshot displays the Visual Studio Code interface. The left sidebar shows a file explorer with a project named 'EXPERIMENT_1' containing several task files (Task_1.exe to Task_8.exe) and source files (Task_1.cpp to Task_8.cpp). The main editor window shows the source code for 'Experiment_1_Task_4.cpp'. The code is as follows:

```
src > Experiment_1_Task_4.cpp > main()
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5
6     char character = 0;
7
8     cout << "Enter a character: ";
9     cin >> character;
10
11     cout << "ASCII value of '" << character << "' is: " << static_cast<int>(character) << endl;
12
13     return 0;
14 }
```

The bottom panel shows the output of the program. The command prompt displays the following text:

```
PS D:\OneDrive\Desktop\Experiment_1> ./bin/Experiment_1_Task_4.exe
Enter a character: 9
ASCII value of '9' is: 57
PS D:\OneDrive\Desktop\Experiment_1>
```

On the right side of the bottom panel, there is a list of open windows, including several 'powershell' instances and 'Experiment_1_Task_4.cpp'. A notification at the bottom right suggests installing the 'Plan agent' for better performance.

Task 5: Integer Division and Remainder Logic

1. Description:

- 1) Define the variables A and B.

```
int A = 0;  
int B = 0;
```

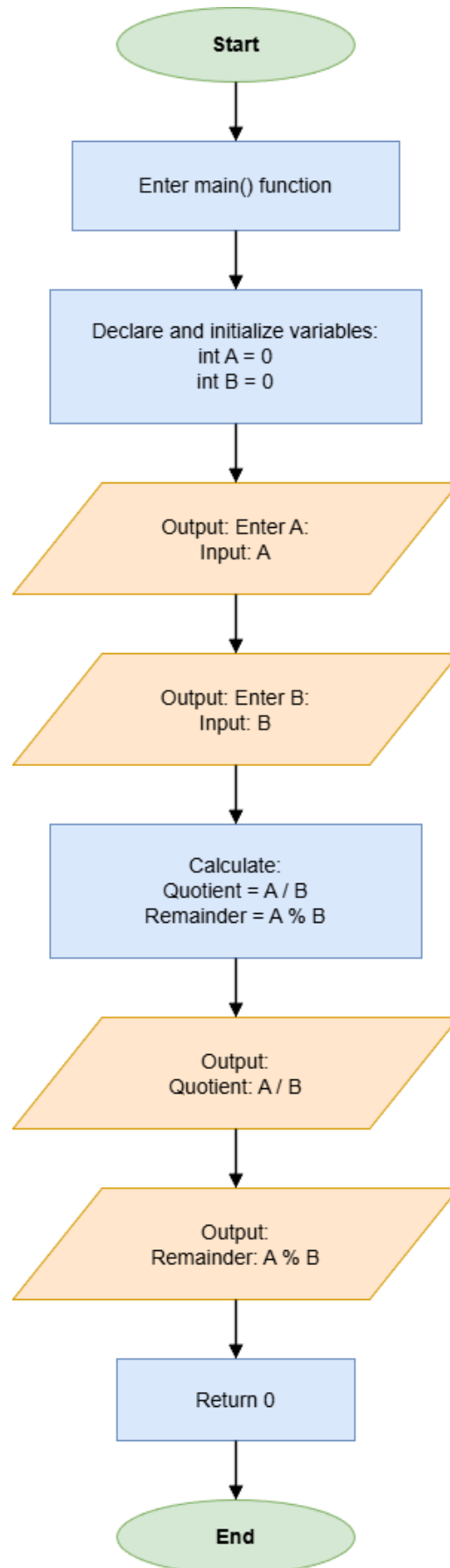
- 2) Input A and B.

```
cout << "Enter A: ";  
cin >> A;  
cout << "Enter B: ";  
cin >> B;
```

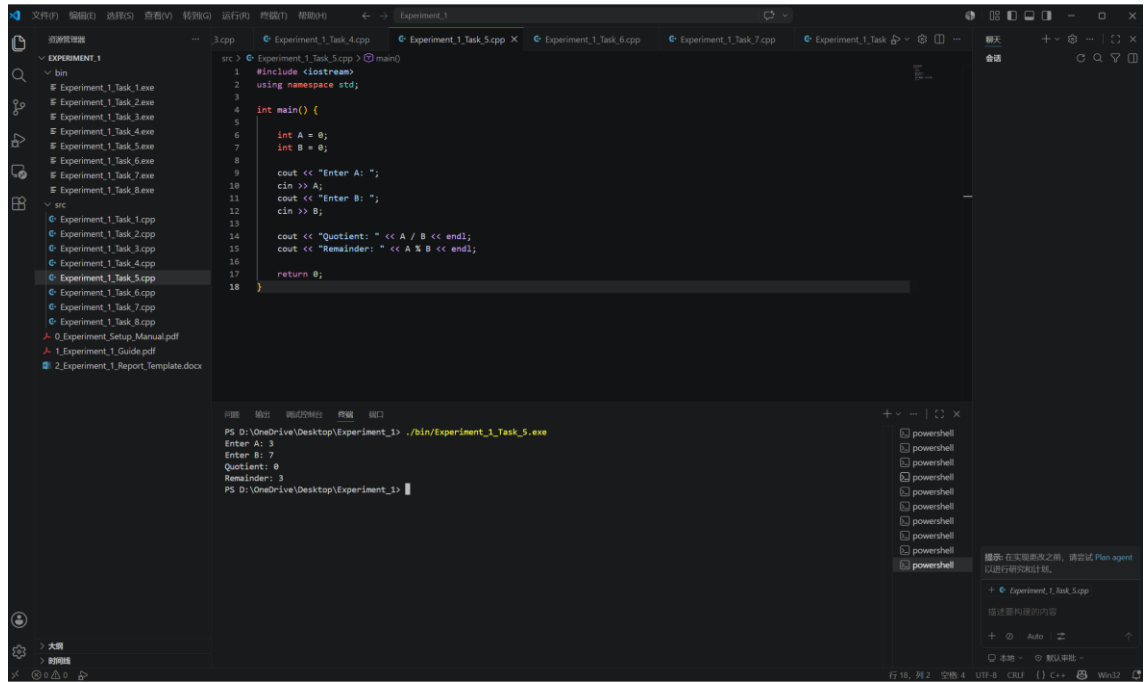
- 3) Print the Quotient and Remainder.

```
cout << "Quotient: " << A / B << endl;  
cout << "Remainder: " << A % B << endl;
```

2. Flow Diagram:



3. Results (Output Screenshot):



The screenshot displays the Visual Studio Code interface. The left sidebar shows a file explorer with a project named 'EXPERIMENT_1' containing subfolders 'bin' and 'src'. The 'src' folder is expanded, showing several C++ files. The main editor window displays the code for 'Experiment_1_Task_5.cpp'. The code is a C++ program that prompts the user to enter two integers, A and B, and then calculates and displays their quotient and remainder. The output window at the bottom shows the program's execution, with the prompt 'Enter A: 3' followed by 'Enter B: 7', and the resulting output 'Quotient: 0' and 'Remainder: 3'. The status bar at the bottom indicates the file is 'Experiment_1_Task_5.cpp' and the language is 'C++'.

```
src > Experiment_1_Task_5.cpp > main()
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5
6     int A = 0;
7     int B = 0;
8
9     cout << "Enter A: ";
10    cin >> A;
11    cout << "Enter B: ";
12    cin >> B;
13
14    cout << "Quotient: " << A / B << endl;
15    cout << "Remainder: " << A % B << endl;
16
17    return 0;
18 }
```

PS D:\OneDrive\Desktop\Experiment_1> ./bin/Experiment_1_Task_5.exe
Enter A: 3
Enter B: 7
Quotient: 0
Remainder: 3
PS D:\OneDrive\Desktop\Experiment_1>

Task 6: Pythagorean Theorem

1. Description:

- 1) Input two sides of a right triangle.

```
int a = 0;
int b = 0;

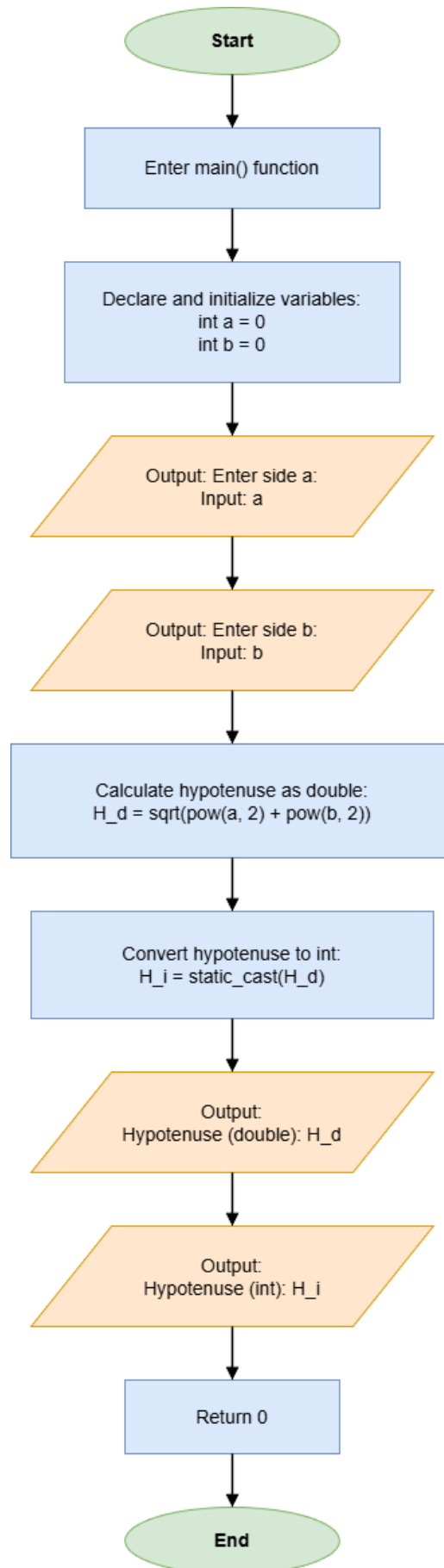
cout << "Enter side a: ";
cin >> a;
cout << "Enter side b: ";
cin >> b;
```

- 2) Calculate the hypotenuse and print it in double and int.

```
double H_d = sqrt(pow(a, 2) + pow(b, 2));
int H_i = static_cast<int>(H_d);

cout << "Hypotenuse (double): " << H_d << endl;
cout << "Hypotenuse (int): " << H_i << endl;
```

2. Flow Diagram:



3. Results (Output Screenshot):

The screenshot displays a C++ IDE with the following components:

- File Explorer (Left):** Shows a project structure with a `bin` directory containing executables `Experiment_1_Task_1.exe` through `Experiment_1_Task_8.exe`, and a `src` directory containing source files `Experiment_1_Task_1.cpp` through `Experiment_1_Task_8.cpp`. It also lists PDF and DOCX files.
- Editor (Center):** Displays the source code for `Experiment_1_Task_6.cpp`. The code includes `<iostream>` and `<cmath>`, uses the `std` namespace, and implements a `main` function that calculates the hypotenuse of a right triangle with sides `a` and `b`.
- Console (Bottom):** Shows the execution output for `./bin/Experiment_1_Task_6.exe`. The user enters `3` for side `a` and `4` for side `b`. The program outputs the hypotenuse as a double (`5`) and as an integer (`5`).
- Terminal (Bottom Right):** A list of PowerShell sessions, with the first one selected.

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4
5
6
7 int main() {
8     int a = 0;
9     int b = 0;
10    cout << "Enter side a: ";
11    cin >> a;
12    cout << "Enter side b: ";
13    cin >> b;
14
15    double H_d = sqrt(pow(a, 2) + pow(b, 2));
16    int H_i = static_cast<int>(H_d);
17
18    cout << "Hypotenuse (double): " << H_d << endl;
19    cout << "Hypotenuse (int): " << H_i << endl;
20
21    return 0;
22 }
```

```
PS D:\OneDrive\Desktop\Experiment_1> ./bin/Experiment_1_Task_6.exe
Enter side a: 3
Enter side b: 4
Hypotenuse (double): 5
Hypotenuse (int): 5
PS D:\OneDrive\Desktop\Experiment_1>
```

Task 7: Integer Overflow Observation

1. Description:

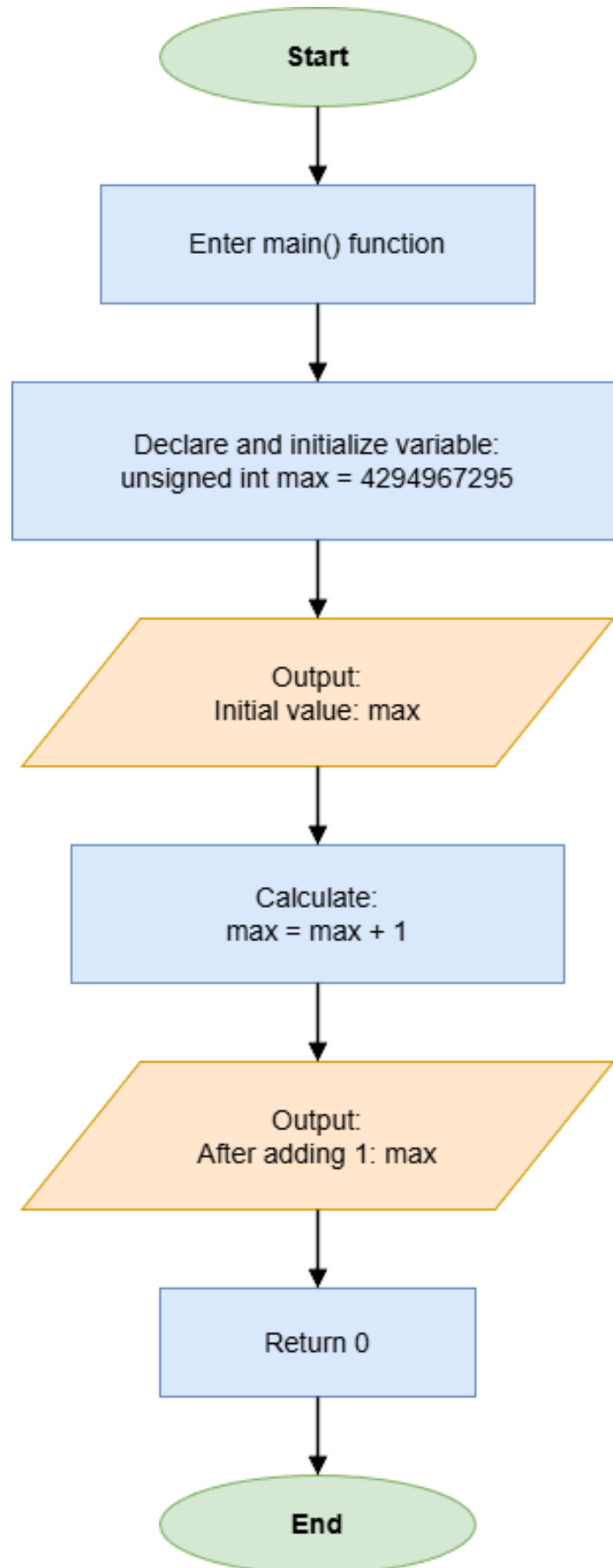
- 1) Initialize an unsigned int with the value 4294967295 and display it.

```
unsigned int max = 4294967295;  
  
cout << "Initial value: " << max << endl;
```

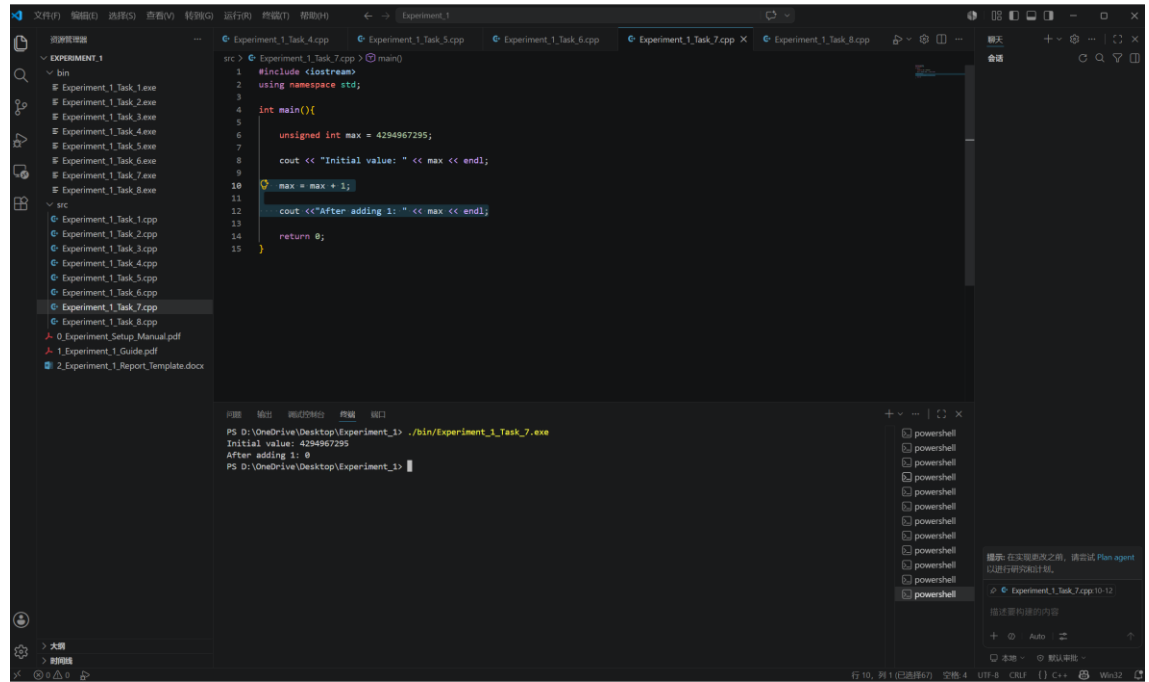
- 2) Add 1 to it and print the result.

```
max = max + 1;  
  
cout << "After adding 1: " << max << endl;
```

2. Flow Diagram:



3. Results (Output Screenshot):



The screenshot displays the Visual Studio Code interface. The left sidebar shows a file explorer with a project named 'EXPERIMENT_1' containing a 'bin' directory with several '.exe' files and a 'src' directory with several '.cpp' files. The main editor window shows a C++ file named 'Experiment_1_Task_7.cpp' with the following code:

```
1 #include <iostream>
2 using namespace std;
3
4 int main(){
5
6     unsigned int max = 4294967295;
7
8     cout << "Initial value: " << max << endl;
9
10    max = max + 1;
11
12    cout << "After adding 1: " << max << endl;
13
14    return 0;
15 }
```

The bottom panel shows the output of the program, which is displayed in a PowerShell terminal window. The output is:

```
PS D:\OneDrive\Desktop\Experiment_1> ./bin/Experiment_1_Task_7.exe
Initial value: 4294967295
After adding 1: 0
PS D:\OneDrive\Desktop\Experiment_1>
```

The right sidebar shows a list of open files, including several 'powershell' files and 'Experiment_1_Task_7.cpp:10-12'. A message at the bottom right suggests trying 'Plan agent' for research and planning.

Task 8: Professional Invoice Formatting

1. Description:

- 1) Input the prices of three items.

```
double price1 = 0;
double price2 = 0;
double price3 = 0;

cout << "Enter prices of three items: ";
cin >> price1 >> price2 >> price3;
```

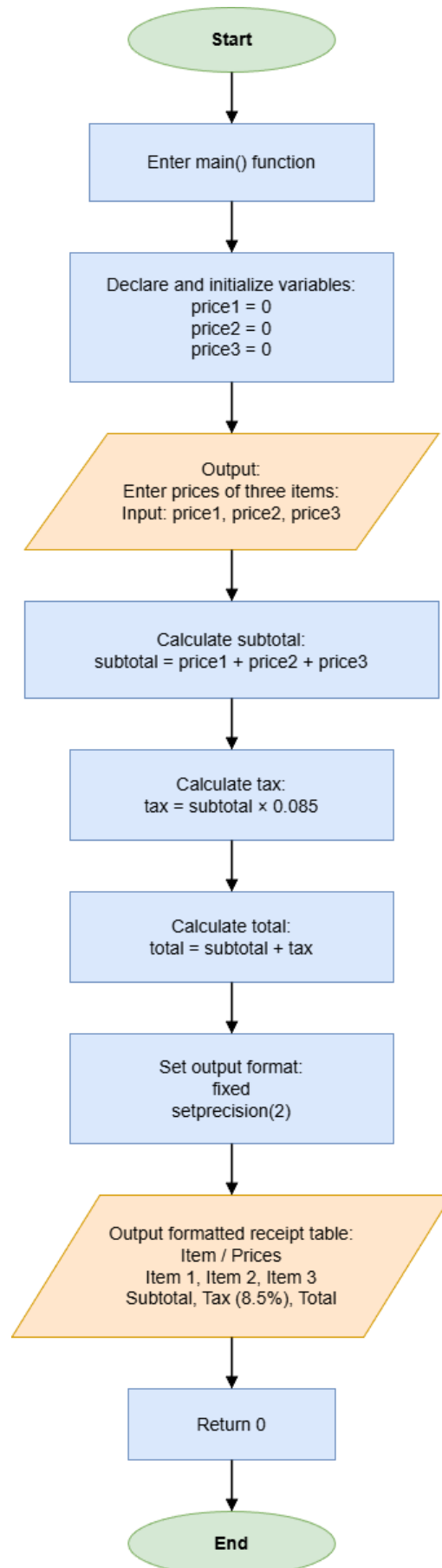
- 2) Calculate subtotal, tax(8.5%), and total.

```
double subtotal = price1 + price2 + price3;
double tax = subtotal * 0.085;
double total = subtotal + tax;
```

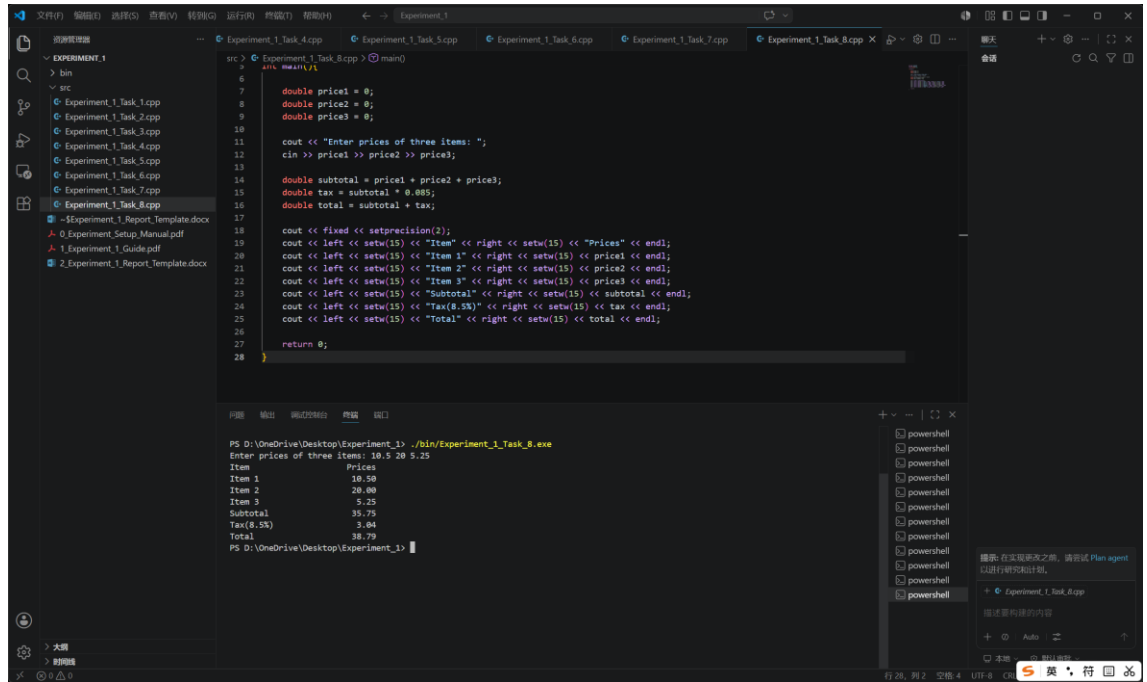
- 3) Format console output using <iomanip>.

```
cout << fixed << setprecision(2);
cout << left << setw(15) << "Item" << right << setw(15) << "Prices"
<< endl;
cout << left << setw(15) << "Item 1" << right << setw(15) << price1
<< endl;
cout << left << setw(15) << "Item 2" << right << setw(15) << price2
<< endl;
cout << left << setw(15) << "Item 3" << right << setw(15) << price3
<< endl;
cout << left << setw(15) << "Subtotal" << right << setw(15) <<
subtotal << endl;
cout << left << setw(15) << "Tax(8.5%)" << right << setw(15) << tax
<< endl;
cout << left << setw(15) << "Total" << right << setw(15) << total <<
endl;
```


2. Flow Diagram:



3. Results (Output Screenshot):



```
src > Experiment_1_Task_8.cpp > main()
6
7     double price1 = 0;
8     double price2 = 0;
9     double price3 = 0;
10
11     cout << "Enter prices of three items: ";
12     cin >> price1 >> price2 >> price3;
13
14     double subtotal = price1 + price2 + price3;
15     double tax = subtotal * 0.085;
16     double total = subtotal + tax;
17
18     cout << fixed << setprecision(2);
19     cout << left << setw(15) << "Item" << right << setw(15) << "Prices" << endl;
20     cout << left << setw(15) << "Item 1" << right << setw(15) << price1 << endl;
21     cout << left << setw(15) << "Item 2" << right << setw(15) << price2 << endl;
22     cout << left << setw(15) << "Item 3" << right << setw(15) << price3 << endl;
23     cout << left << setw(15) << "Subtotal" << right << setw(15) << subtotal << endl;
24     cout << left << setw(15) << "Tax(8.5%)" << right << setw(15) << tax << endl;
25     cout << left << setw(15) << "Total" << right << setw(15) << total << endl;
26
27     return 0;
28 }
```

PS D:\OneDrive\Desktop\Experiment_1> .\bin\Experiment_1_Task_8.exe
Enter prices of three items: 10.5 20 5.25
Item Prices
Item 1 10.50
Item 2 20.00
Item 3 5.25
Subtotal 35.75
Tax(8.5%) 3.04
Total 38.79
PS D:\OneDrive\Desktop\Experiment_1>

IV. Extended Questions

1. In Task 5, when $A = 3$ and $B = 7$, why is the quotient 0?

Because A and B are both integers, C++ performs integer division. The mathematical result of $3 / 7$ is approximately 0.4286, but integer division discards the fractional part instead of rounding it. Therefore, the quotient is 0 and the remainder is 3.

2. In Task 7, why does the value become 0 after adding 1 to 4294967295?

The value 4294967295 is the maximum value of a 32-bit unsigned int. Adding 1 produces 4294967296 mathematically, but this value is outside the storage range. Unsigned integer arithmetic wraps around modulo 2^{32} , so the result returns to 0.

V. Summary and Reflections

In this experiment, I practiced the basic knowledge of C++ programming, including program structure, comments, variables, constants, data types, operators, type conversion, mathematical functions, and formatted output. Through the eight tasks, I became more familiar with writing, compiling, running, and checking simple C++ programs.

In Task 1, I learned the basic structure of a C++ program and practiced using comments to make the code clearer.

In Task 2, I used constants and unsigned integer variables to calculate the total value of coins, which helped me understand the difference between constants and variables.

In Task 3, I used the `sizeof` operator to observe the memory size of different data types.

In Task 4, I converted a character to its ASCII value using `static_cast<int>()`, which helped me understand character storage and type conversion.

In Task 5, I practiced integer division and the remainder operator. I learned that integer division discards the fractional part.

In Task 6, I used the Pythagorean theorem with `sqrt()` and `pow()` to calculate the hypotenuse, and observed the difference between double and int results.

In Task 7, I observed unsigned integer overflow, which showed that computer arithmetic is limited by data type ranges.

In Task 8, I used `<iomanip>` to format an invoice output, making the result clearer and more professional.

During the experiment, the main difficulties I encountered were how to compile .cpp files correctly and how to use `<iomanip>` to control decimal places and format the output professionally. At first, I was not familiar with the compiling process in Visual Studio Code, so I needed to learn how to open the terminal, use the correct command, and run the generated program. In Task 8, I also spent time learning how to use `fixed`, `setprecision()`, `setw()`, `left`, and `right` to make the invoice output neat and readable. By testing the programs several times and asking AI, I gradually solved these problems. Overall, this experiment helped me build a basic foundation in C++ programming and improved my ability to write and run simple programs.